ALENDRONIC ACID

Strict administration rules

Weekly dose

CHEMICAL NAME (4-amino-1-hydroxy-1-phosphonobutyl)phosphonic acid (sodium alendronate trihydrate)

CLASS Bisphosphonate (bone resorption inhibitor)

TRADE NAMES Fosamax · Binosto · Fosavance (with vitamin D) · Alendronate

THERAPEUTIC USES / INDICATIONS

- Postmenopausal osteoporosis; treatment and prevention of fractures.
- Osteoporosis in men; glucocorticoid-induced osteoporosis.
- Paget's disease of bone.

ADMINISTRATION

- **PO** on waking, on an empty stomach, with a full glass (200 mL) of plain water.
- Remain upright (sitting/standing) for at least 30 min and until after first food.
- Wait 30 min before food, drink or other medicines.
- Do not chew or suck tablet; not at bedtime.

DOSAGES

- **Osteoporosis:** 70 mg once weekly (or 10 mg daily).
- **Steroid-induced prevention:** 70 mg weekly (or 5–10 mg daily).
- **Paget's:** 40 mg daily for 6 months.
- Ensure calcium and vitamin D adequacy; review after 5 years (drug holiday).

PHARMACOKINETICS

- Bioavailability <1% (↓↓ by food, calcium, coffee).
- Half absorbed drug binds to bone; not metabolised.
- Rest excreted in urine. Bone **t½:** >10 years.

PHARMACODYNAMICS

- Binds **hydroxyapatite** in bone; inhibits **osteoclast** activity (farnesyl pyrophosphate synthase) → ↓ bone resorption → ↑ bone density, ↓ fractures.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dyspepsia, abdominal pain, acid regurgitation.
- Musculoskeletal pain.
- Headache.

ADVERSE EFFECTS (SERIOUS)

- Oesophagitis, oesophageal ulcers/strictures (incorrect administration).
- Osteonecrosis of the jaw (dental extractions, cancer, poor oral hygiene).
- Atypical femoral fractures (long-term).
- Hypocalcaemia; osteonecrosis of external auditory canal.

CONTRAINDICATIONS & CAUTIONS

- Oesophageal abnormalities delaying emptying (stricture, achalasia).
- Inability to stand or sit upright for 30 min.
- Hypocalcaemia (correct first).
- eGFR <35 mL/min.

KEY INTERACTIONS

- **Calcium, iron, antacids, milk, food:** ↓ absorption — separate.
- **NSAIDs:** ↑ GI irritation.

NURSING MANAGEMENT

- Teach and check the administration routine (weekly same day).
- Check calcium and vitamin D before starting; renal function.
- Dental check before starting; good oral hygiene.
- Report thigh/groin pain (atypical fracture).

PATIENT EDUCATION

- Take on waking with a full glass of plain water; stay upright and wait 30 min before breakfast, drinks or medicines.
- Stop and report chest pain, difficulty or pain swallowing, heartburn.
- Tell your dentist; report jaw pain or loose teeth.
- Take calcium/vitamin D at a different time of day.

KEY POINTS

- Bisphosphonate — inhibits osteoclasts.
- Oesophagitis; jaw osteonecrosis; atypical fractures.
- 70 mg weekly, strict empty-stomach, upright 30 min.
- Correct Ca/vitamin D; eGFR ≥35.